# SABRINA HENG WEI QI (A23CS0265)
# SECP3623 – DATABASE PROGRAMMING
# SEM I – 2025/2026
# MongoDB(shopdb.orders)

**Instruction:**

Insert the following data into the shopdb database and the orders collection.

```
use shopdb

db.orders.insertMany([
 {
  order_id: "ORD101",
  customer: "Ali Ahmad",
  product: "Laptop",
  quantity: 1,
  price: 3500,
  status: "Completed",
  order_date: new Date("2025-01-05")
 },
 {
  order_id: "ORD102",
  customer: "Siti Aminah",
  product: "Smartphone",
  quantity: 2,
  price: 2800,
  status: "Pending",
  order_date: new Date("2025-01-06")
 },
 {
  order_id: "ORD103",
  customer: "John Lee",
  product: "Tablet",
  quantity: 1,
  price: 1800,
  status: "Completed",
  order_date: new Date("2025-01-07")
 },
 {
  order_id: "ORD104",
  customer: "Nur Aisyah",
  product: "Laptop",
  quantity: 1,
  price: 3700,
  status: "Cancelled",
  order_date: new Date("2025-01-08")
```

```
 },
 {
  order_id: "ORD105",
  customer: "Ahmad Zaki",
  product: "Headphones",
  quantity: 3,
  price: 450,
  status: "Completed",
  order_date: new Date("2025-01-09")
 },
 {
  order_id: "ORD106",
  customer: "Farah Izzati",
  product: "Smartwatch",
  quantity: 2,
  price: 1200,
  status: "Pending",
  order_date: new Date("2025-01-10")
 }
])
```

a) Write a MongoDB command to display all orders with status "Completed".

db.orders.find ( { status : "Completed" })

Output:

```
  _id: ObjectId('695fa103553c77fd40a3042c'),
  order_id: 'ORD101',
  customer: 'Ali Ahmad',
  product: 'Laptop',
  quantity: 1,
  price: 3500,
  status: 'Completed',
  order_date: 2025-01-05T00:00:00.000Z
}
{
  _id: ObjectId('695fa103553c77fd40a3042e'),
  order_id: 'ORD103',
  customer: 'John Lee',
  product: 'Tablet',
  quantity: 1,
  price: 1800,
  status: 'Completed',
  order_date: 2025-01-07T00:00:00.000Z
}
{
  _id: ObjectId('695fa103553c77fd40a30430'),
  order_id: 'ORD105',
  customer: 'Ahmad Zaki',
  product: 'Headphones',
  quantity: 3,
  price: 450,
  status: 'Completed',
  order_date: 2025-01-09T00:00:00.000Z
```

b) Write a MongoDB command to display all orders with price greater than RM2000.

db.orders.find({ price : { $gt : 2000 }})

Output:

```
  _id: ObjectId('695fa103553c77fd40a3042c'),
  order_id: 'ORD101',
  customer: 'Ali Ahmad',
  product: 'Laptop',
  quantity: 1,
  price: 3500,
  status: 'Completed',
  order_date: 2025-01-05T00:00:00.000Z
}
{
  _id: ObjectId('695fa103553c77fd40a3042d'),
  order_id: 'ORD102',
  customer: 'Siti Aminah',
  product: 'Smartphone',
  quantity: 2,
  price: 2800,
  status: 'Pending',
  order_date: 2025-01-06T00:00:00.000Z
}
{
  _id: ObjectId('695fa103553c77fd40a3042f'),
  order_id: 'ORD104',
  customer: 'Nur Aisyah',
  product: 'Laptop',
  quantity: 1,
  price: 3700,
  status: 'Cancelled',
  order_date: 2025-01-08T00:00:00.000Z
```

c) Write a MongoDB command to display all orders sorted by price in descending order.

db.orders.find().sort({ price : -1 })

Output:

```
  _id: ObjectId('695fa103553c77fd40a3042f'),
  order_id: 'ORD104',
  customer: 'Nur Aisyah',
  product: 'Laptop',
  quantity: 1,
  price: 3700,
  status: 'Cancelled',
  order_date: 2025-01-08T00:00:00.000Z
}
{
  _id: ObjectId('695fa103553c77fd40a3042c'),
  order_id: 'ORD101',
  customer: 'Ali Ahmad',
  product: 'Laptop',
  quantity: 1,
  price: 3500,
  status: 'Completed',
  order_date: 2025-01-05T00:00:00.000Z
}
{
  _id: ObjectId('695fa103553c77fd40a3042d'),
  order_id: 'ORD102',
  customer: 'Siti Aminah',
  product: 'Smartphone',
  quantity: 2,
  price: 2800,
  status: 'Pending',
  order_date: 2025-01-06T00:00:00.000Z
```

```
  _id: ObjectId('695fa103553c77fd40a3042e'),
  order_id: 'ORD103',
  customer: 'John Lee',
  product: 'Tablet',
  quantity: 1,
  price: 1800,
  status: 'Completed',
  order_date: 2025-01-07T00:00:00.000Z
}
{
  _id: ObjectId('695fa103553c77fd40a30431'),
  order_id: 'ORD106',
  customer: 'Farah Izzati',
  product: 'Smartwatch',
  quantity: 2,
  price: 1200,
  status: 'Pending',
  order_date: 2025-01-10T00:00:00.000Z
}
{
  _id: ObjectId('695fa103553c77fd40a30430'),
  order_id: 'ORD105',
  customer: 'Ahmad Zaki',
  product: 'Headphones',
  quantity: 3,
  price: 450,
  status: 'Completed',
  order_date: 2025-01-09T00:00:00.000Z
```

d) Write a MongoDB command to display all "Pending" orders sorted by order_date (latest first).

```
db.orders.find ( { status : "Pending" } ). sort ( { order_date : -1 } )
```

Output:

```
< {
  _id: ObjectId('695fa103553c77fd40a30431'),
  order_id: 'ORD106',
  customer: 'Farah Izzati',
  product: 'Smartwatch',
  quantity: 2,
  price: 1200,
  status: 'Pending',
  order_date: 2025-01-10T00:00:00.000Z
}
{
  _id: ObjectId('695fa103553c77fd40a3042d'),
  order_id: 'ORD102',
  customer: 'Siti Aminah',
  product: 'Smartphone',
  quantity: 2,
  price: 2800,
  status: 'Pending',
  order_date: 2025-01-06T00:00:00.000Z
}
```

e) Using aggregation, calculate the total sales amount (price) for each product.

db. orders. aggregate ([
{ group : {_id : "$ product" , total_sales_amount : {$sum : "$ price"}}}
])

Output:

```
< {
    _id: 'Smartwatch',
    total_sales_amount: 1200
  }
  {
    _id: 'Smartphone',
    total_sales_amount: 2800
  }
  {
    _id: 'Laptop',
    total_sales_amount: 7200
  }
  {
    _id: 'Tablet',
    total_sales_amount: 1800
  }
  {
    _id: 'Headphones',
    total_sales_amount: 450
  }
```

f) Using aggregation, calculate the total sales amount for orders with status "Completed".

db. orders. aggregate([
{ $ match : { status : "Completed" }},
{ $ group : { _id : "Total Completed Sales Amount", total_sales_amount : { $ sum : "$price" }}}
])

Output:

```
< {
    _id: 'Total Completed Sales Amount',
    total_sales_amount: 5750
  }
```

g) Write a MongoDB command to update the status of order ORD102 from "Pending" to "Completed".

db. orders. updateOne ( { order_id : "ORD 102"},
{ $ set : { status : "Completed }})

Output:

```
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
```

```
_id: ObjectId('695fa103553c77fd40a3042d')
order_id : "ORD102"
customer : "Siti Aminah"
product : "Smartphone"
quantity : 2
price : 2800
status : "Completed"
order_date : 2025-01-06T00:00:00.000+00:00
```

h) Write a MongoDB command to increase the quantity of order ORD105 by 1.

db.orders.updateOne( {order_id : "ORD105"} ,
{ $ inc : {quantity :1}}
)

Output:

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
_id: ObjectId('695fa103553c77fd40a30430')
order_id : "ORD105"
customer : "Ahmad Zaki"
product : "Headphones"
quantity : 4
price : 450
status : "Completed"
order_date : 2025-01-09T00:00:00.000+00:00
```

i) Write a MongoDB command to delete the order with order_id "ORD104".

db.orders.deleteOne ( { order_id : "ORD104"} )

Output:

```
{
  acknowledged: true,
  deletedCount: 1
}
```

j) Write a MongoDB command to delete all orders with status "Cancelled".

db.orders.deleteMany ( { status : "Cancelled"} )

Output:

```
{
  acknowledged: true,
  deletedCount: 0
}
```